

Hydrocarbon Potential of the British Isles and North Sea

Executive Summary: The UK Continental Shelf (UKCS) remains a major hydrocarbon province: ~47.7 billion barrels of oil equivalent (boe) have been produced to date [70†L199-L202] . Only about 2.9 bn boe of 2P (proved+probable) reserves remain (~19% of cumulative production) [70†L199-L202] . Key petroleum systems include Jurassic sources (Kimmeridge Clay) charging multi-era reservoirs in the central North Sea, and Permo-Triassic Rotliegend gas plays in the Southern North Sea [29†L248-L256] [70†L199-L202] . Major offshore basins (Central, Northern and Southern North Sea, West of Shetland, Irish Sea) each host distinct plays (see below). Onshore UK (Wessex/Weald, East Midlands, Cheshire/Lancashire) also has Jurassic/Triassic and Carboniferous systems generating oil and gas [58†L37-L40] [59†L7-L16] . Exploration has matured in proven plays, shifting focus to underexplored deep and Paleozoic targets (e.g. Carboniferous plays on the Mid-North Sea High and Orcadian basins [28†L225-L232]). Data availability is high: BGS's Offshore GeoIndex and NSTA's open-data portals provide extensive seismic and well data [28†L269-L274] . Technical risks include HPHT overpressure in deep Jurassic/Permian traps [45†L211-L219] , sour-gas in some deep fields, and variable seal/reservoir quality (though Paleocene fan sands can have >25% porosity [7†L1494-L1502]). Regulatory/economic constraints are stiff: **no new UK O&G licences** will be issued under current policy [70†L210-L213] , decommissioning costs are very high (£2.4 bn spent in 2024; £27 bn forecast by 2032 [49†L197-L205]), and upstream emissions (CO₂ venting) will face carbon pricing from 2025 [52†L97-L100] . Priority targets combine attractive geology with existing infrastructure: e.g. shallow Carboniferous/Permian leads near pipelines (reducing capex) and deep-water stratigraphic traps on basin flanks. This report reviews geology, fields, prospects and constraints basin-by-basin, and recommends high-impact plays with risk/reward.

Petroleum Geology & Systems

The UK region spans multiple petroleum provinces. **North Sea – Oil Province:** Jurassic source rocks (Kimmeridge/Oxford Clay) charge overlying reservoirs from Late Jurassic to Paleogene. For example, the giant Forties Field (discovered 1970) produces oil from Paleocene turbidite fan reservoirs (Forties Formation) sealed by Kimmeridge Clay [29†L248-L256] . The Brent/Ninian complex (Brent Group sandstones, Upper Jurassic, ~3–4 km depth) is charged by the same Jurassic shales. **North Sea – Gas Province:** Permian Rotliegend aeolian sandstones (3–4 km depth) form prolific gas reservoirs (Indefatigable, Leman fields) with Zechstein evaporites as seals. **Atlantic Margin (West of Shetland):** Extensive Paleocene-age fan and slope complexes (e.g. Foinaven, Schiehallion) lie over fractured Paleozoic basement. These deep-water fans have excellent reservoir quality (porosity >25%, permeability >100 mD where <2.5 km burial) but rely on subtle stratigraphic pinchouts [7†L1494-L1502] . Jurassic Kimmeridge Clay also provides source here. **Southern North Sea:** Hosts gas in Permian dune sands (Rotliegend) and Carboniferous basement flanks; e.g. Breagh Field (1999) has 2.8 tcf gas in Carboniferous sandstone [28†L225-L233] .

Onshore UK: Southern England (Wessex, Weald) has a Jurassic petroleum system (oil from Kimmeridge/Oxford Clay migrated into Jurassic-age reservoirs) [58†L37-L40] . Wytch Farm (Dorset) – Western Europe's largest onshore oilfield – produces Jurassic oil (Bridport/Portland sands) from Jurassic fault blocks [57†L85-L88] . The East Midlands/Pennine province is exclusively Carboniferous: gas and minor

oil in Pennine coal measures and sandstones 【59†L7-L16】 . The Irish Sea (East Irish Sea Basin and adjacent Cheshire/Lancashire) is Permo-Triassic; mid-Carboniferous Namurian shales (e.g. Holywell/Bowland Shales) act as source, charging Triassic sandstones (Helsby/Ormskirk) – e.g. the Formby oilfield overlies Bowland shale 【60†L132-L140】 . A mapping study (NSTA/BGS “21CXR”) has recently highlighted potential Devonian–Carboniferous plays across the Mid-NS High, Orcadian, and Irish Sea basins, calibrated to Breagh/Western Irish Sea analogues 【28†L225-L233】 .

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flowchart LR
    NS_Oil[North Sea Oil Province] --> Jurassic_Play[Jurassic syn-rift plays  
(source: Kimmeridge Clay)]
    NS_Oil --> PostRift_Play[Post-rift Cretaceous–Paleogene plays]
    NS_Gas[North Sea Gas Province] --> Permian_Play[Permian Rotliegend gas plays]
    W_Shetland[West of Shetland (Atlantic Margin)] --> Paleocene_Fan[Paleocene fan reservoirs]
    IR_Sea[East Irish Sea Basin] --> Carboniferous_Play[Mid-Carboniferous source]
    Onshore[Onshore UK Basins] --> Weald_Wessex[Jurassic (Weald/Wessex oil)]
    Onshore --> Midlands_Carbon[Carboniferous (East Midlands oil/gas)]

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Diagram: Petroleum plays by region/basin (arrows indicate dominant source-reservoir relationships).

Basins and Provinces

We review basins basin-by-basin (UK sectors plus adjacent):

- **Central North Sea (CNS):** Major oil province (Brent, Forties, Ninian, Buzzard fields). Reservoirs span Middle Jurassic Brent Group to Paleocene turbidite fans 【29†L248-L256】 . Jurassic Kimmeridge Clay is the chief source. Typical traps are tilted fault blocks and rollovers. *Exploration:* Mature; new focus on deep re-interpretation and HPHT plays (Culzean) with Jurassic/Lower Cretaceous targets 【45†L211-L219】 .
- **Northern North Sea (East Shetland):** Prolific Jurassic/Permian oil (e.g. Britannia, Clair) and gas (J-Block). Shallow Jurassic clastics (Montrose/Scapa Sandstone) and Permian sandstones produce oil. Tight structural traps (basin inversion plays) are common. *Exploration:* 3D seismic has rejuvenated deeper (Triassic) targets and prograding shoreface plays; Paleozoic reservoirs on high structures are underplayed.
- **Northern Moray Firth:** Chalk and Jurassic sandstone gas fields (e.g. Captain, Erskine). Jurassic mudstones (Source: Kimmeridge) charge Triassic sandstone reservoirs. Structural inversion blocks on basin edges hold small pools. *Exploration:* Mature, minor remaining prospects in Paleocene fans and in chalk traps.
- **Southern North Sea (SNS):** Predominantly Permian gas province (fields like West Sole, Indefatigable, Leman, Trent). Permian Rotliegend aeolian sands (2–3 km) are reservoir; sealed by Zechstein evaporites. Some Triassic (Sherwood) gas in East Ireland Sea sector. *Exploration:* Largely mature; focus has shifted to satellite tie-backs and strat traps, or exploration in Irish Sea Permo-Triassic (e.g. Breagh, Hamilton).

- **West of Shetland (Atlantic Margin):** Deepwater frontier (Foinaven, Schiehallion oil fields). Thick Paleocene fan/slope systems (2–3 km, high quality) have been proven [7†L1494-L1502] . Jurassic source rocks charge these plays. Complex salt/structural deformation (e.g. Ymir oil) offers high-risk exploration. *Exploration:* Growing interest in deeper (“Deepwater” license-round), and new Triassic/Permian targets (e.g. Duva gas).
- **Rockall Basin (UK side):** Deep basinal Permo-Trias fill (est. ~8–9 km thick). Highly underexplored; on the UK side the only discovery is the two-well Dooish gas (1992). Source potential uncertain (maybe Paleozoic). *Exploration:* Remains frontier; interest in seismic imaging of potential Cretaceous/Paleocene sandstones.
- **Irish Sea Basins:** Includes Cardigan Bay (Celtic Sea) and East Irish Sea. *Cardigan Bay (Irish Sea):* A Permo-Triassic half-graben near Wales. Jurassic source is mature (oil shows, Dragon gas discovery [71†L1201-L1204]). The Dragon oil/gas well (mid-Jurassic reservoir) proved the system. Small stratigraphic traps exist. *East Irish Sea:* Permo-Triassic rift basin on UK–Irish shelf. Carboniferous shales (Namurian) source oil into Triassic sandstone reservoirs; Formby (W Lancashire) and Hamilton oilfields are onshore extensions [60†L78-L85] . Offshore gas fields (E. Irish Sea) are from Mid Carboniferous source. *Exploration:* Moderate activity; license rounds (29th, 30th) targeted these Paleozoic plays and Permo-Triassic traps.
- **UK Onshore Basins:**
 - *Wessex/Channel:* Jurassic rift grabens (Wessex and Weald basins). Source rocks (Kimmeridge/ Oxford Clay) have generated oil [58†L37-L40] . Wytch Farm (Jurassic Bridport and Sherwood reservoirs) exemplifies the play [57†L85-L88] . Faulted blocks at inversion highs (Cranborne, Purbeck) trap oil. Additional small Jurassic and Triassic fields (Wareham, Kimmeridge Oilfield) occur.
 - *Weald:* Adjacent to Wessex; similar Jurassic play with multiple small oilfields (e.g. Horse Hill). Kimmeridge mature, charging Bridport limestones.
 - *East Midlands/Cleveland:* Pennine Carboniferous synclines and blocks (e.g. Gainsborough Trough) host gas fields. Hydrocarbons are in Carboniferous (gas from coal measures, oils in Namurian reef facies). Prospectivity is low to moderate; few discoveries (Melton Mowbray) but significant shale (Bowland) source is present.
 - *Lancashire/Cheshire:* Onshore extension of E. Irish Sea. Permo-Triassic sandstone (Helsby, Sherwood) reservoirs (Elswick Gasfield) with Carboniferous source (Holywell/Bowland) and Permian seal. Minor gas has been produced (Rossendale, Aughton).
 - *Scotland – Midland Valley:* Deep Paleozoic basin under parts of Scotland. Dinantian oil shales (subject to study) with Carboniferous sand reservoirs on basement highs are theoretical plays.
 - *Northern Ireland:* Antrim Basalt Gas (not industry-scale), Lough Neagh basin. Underexplored; some shale gas/heavy oil potential.
 - *Other (Wales, etc):* Minor fields in Wales; offshore (UK) Celtic Sea beneath Cardigan Bay (Dragon).

Proven Fields and Production

Major UK fields (oil and gas) are summarized in Table 1. These illustrate the diversity of reservoirs and trap styles. For example, the Forties Field (Central NS) was discovered in 1970 and produces from Paleocene turbidites [29†L248-L256] . Wytch Farm (onshore Wessex) was found in 1973 and is the largest UK onshore oilfield (Jurassic reservoirs) [57†L85-L88] . The Breagh gas field (SNS) discovered 1999 yields 2.8 tcf from Carboniferous sands [28†L225-L233] . (Production histories for many smaller fields, and latest reserves, are reported by NSTA's Energy Portal.) These fields, and others like Brent,

Ninian, Piper (CNS oil) or Morecambe Bay, Indefatigable (SNS gas), were found mostly in 1970s–90s by large companies [68†L123-L126] . Cumulative UKCS production peaked in 1999 and is declining; stabilizing production relies on late-life and new developments (Table 2).

Field	Basin/ Block	Disc.	2P Reserves	Cum. Production	Reservoir (depth)	Trap Style
Forties	Central NS (13/30)	1970	$\sim 2.6 \times 10^9$ bbl 【29†L248- L256】	$\sim 3 \times 10^9$ bbl oil	Paleocene turbidite (Forties Fm, ~2500 m)	Tilted fault block
Wytch Farm	Wessex (SY/ Dorset)	1973	$\sim 0.5 \times 10^9$ bbl 【57†L85-L88】	$\sim 0.5 \times 10^9$ bbl oil	Jurassic (Bridport & Sherwood, ~1500 m)	Horst block (faulted)
Breagh (gas)	SNS (49/19)	1999	79×10^9 m ³ gas 【28†L225- L233】	$\approx ?$ (UK's LNG export)	Carboniferous/ Triassic (~2000 m)	Tilted fault block
Buzzard	Northern NS (23/17)	2001	$\sim 0.45 \times 10^9$ bbl	$\sim 0.2 \times 10^9$ bbl oil	Paleocene sand (Flett Fm, ~3000 m)	Structural high
Morecambe Bay	SNS (42/7)	1974	$\sim 310 \times 10^9$ m ³ gas	$\sim 280 \times 10^9$ m ³ gas	Triassic Sherwood/ Permian (~2 km)	Roll-over anticlinal

Table 1. Examples of major UK fields. (2P reserves and production are indicative; sources: NSTA/BGS.)

Table 2 (below) shows UKCS historical output. Production of oil and gas peaked around 1999 (~4.5 MMboe/day) and has since declined, with a recent uptick from new developments [68†L125-L132] . In 2023, UKCS production was ~1.4 MMboe/day (60% oil, 40% net gas) [68†L129-L132] [68†L210-L218] .

Year	UKCS Oil (mmbbl)	UKCS Gas (bcm)	Total (MMboe)	Comments
1999	~153	~49	~1640	Peak production
2014	~58	~35	~470	Historical low
2022	~46	~21	~430	Recent output
2023	~42	~19	~385	Sources: NSTA/DESNZ [68†L125- L132]

Table 2. UKCS production of oil and gas (selected years).

Exploration History & Prospectivity

Since the first North Sea wells in the 1960s, UKCS exploration has matured. Early large discoveries gave way to smaller finds and re-developments. The 2010s saw renewed activity in frontier plays (e.g. Mariner oil 2014, Felix gas 2017 in SNS). However, recent licensing rounds have shrunk (30th round in 2016, 32nd in 2018; no new O&G rounds are now planned [70†L210-L213]). Key recent exploration insights include the NSTA/BGS 21CXRm projects: these have mapped under-explored Devonian/Carboniferous source-reservoir plays. For example, 21CXRm identified Carboniferous sandstone “hotspots” beneath the Mid-North Sea High and Orcadian areas, analogous to the Breagh system [28†L225-L233] . It also confirmed that Namurian/Holywell shales charge the West of Shetland (e.g. Franklin-Hebrides play) and East Irish Sea basins. As a result, remaining prospectivity now focuses on: (1) **Paleozoic basins** – deep Carboniferous/Devonian targets (e.g. MNSH faults, North Celtic Sea), (2) **Mesozoic strat traps** – subtle tilted fault blocks or pinchouts in West Shetland and deep CNS, and (3) **Unconventional/Heavy oils** – low effort so far. Under-explored regions include the UK Rockall basin and Zechstein reefal plays, which saw little drilling.

Seismic & Well Data Sources

The UK’s subsurface data are well catalogued. Key sources include the BGS **Offshore GeoIndex** (mapapps2.bgs.ac.uk) and the NSTA/OGA **open data portal** (UK National Data Repository via ArcGIS), which offer interactive access to tens of thousands of seismic lines and well logs [28†L269-L274] . Onshore seismic/well archives (e.g. UKOGL – Onshore Geophysical Library) provide nearly complete coverage of UK onshore exploration. Recent studies (e.g. 21CXRm) have released interpreted surfaces and well pressure data (194 wells in CNS) to the public. NSTA’s **Energy Portal** also supplies field and well production datasets. In short, nearly all pre-2020 UK seismic and well data are publicly available (subject to license), enabling new prospect mapping.

Structural/Stratigraphic Framework (Maps & Figures)

Key structural maps outline major highs and basins (e.g. Mid-North Sea High, Viking Graben, Sole Pit Trough, Orcadian High, Lusitanian Basin). Stratigraphic cross-sections illustrate reservoir-seal stacks (e.g. Jurassic Kimmeridge source under Brent sand reservoir in CNS; or Triassic Sherwood sand under Zechstein salt in SNS). For example, the simplified basin-play diagram above encapsulates the relationships. (High-resolution GIS maps of licensing, fields, and geology are available from NSTA and BGS.)

flowchart TD

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A[Identify basins & play concepts] --> B[Gather seismic/well data]
B --> C{Quality of data?}
C -- High --> D[Map existing prospects/leads]
C -- Low --> E[Acquire/new seismic surveys]
D --> F[Screen technical risks (HPHT, sour, seal)]
E --> F
F --> G[Assess policy/econ constraints (licences, pricing)]
G --> H[Rank targets by geology vs cost]
H --> I[Select drill candidates]
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Figure: Decision-flow for targeting exploration leads (mermaid diagram).

Technical Risks and Uncertainties

Drilling in the UKCS entails several hazards. **Overpressure and HPHT:** The Central Graben and W. Shetland deep plays are known for over-pressured Jurassic/Triassic reservoirs. An NSTA study confirms a major Jurassic–Triassic pressure cell in the Central North Sea [45†L211-L219] . Wells here can exceed 600 bar and >150°C, raising drilling risk. **Sour gas:** H₂S is rare in UK fields but can occur (e.g. Cyclone Field). Only limited reservoirs (mostly Carboniferous/Pennine gas) have high H₂S. **Reservoir quality:** Most UK reservoirs are excellent (high-porosity sandstones [7†L1494-L1502]), but diagenesis can degrade quality at depth. Shallow Paleocene fans, for example, have outstanding porosity [7†L1494-L1502] , whereas deeply-buried Permian might be tight. **Seal integrity:** Many UK traps rely on regional clay seals (Kimmeridge, Zechstein) which generally cap well, but fault seal risk exists on reactivated structures. **Geomechanics:** Abundant faults and rotated blocks require careful seismic interpretation to avoid drilling non-sealed pockets.

Regulatory, Economic & Environmental Constraints

A number of external constraints heavily influence drilling decisions. **Licensing:** Under current UK policy *no new oil/gas licences will be issued* [70†L210-L213] . Only existing licenses (and associated remaining leads) can be drilled unless policy changes. **Decommissioning:** The basin faces a huge decommissioning burden; in 2024 operators spent £2.4 bn on decommissioning, with £27 bn forecast by 2032 [49†L197-L205] . These costs influence project economics (tie-backs are favored). **Carbon pricing:** From 2025, vented CO₂ from gas processing (sweetening) will be charged under UK ETS [52†L97-L100] , modestly increasing OPEX. **Protected Areas:** Several Marine Protected Areas (MPAs) and offshore wind zones overlap licence blocks; environmental consents can restrict activities. (No-go zones exist for reefs, bird/SPAs). **Markets/Economics:** Oil and gas price volatility remains a risk. Global carbon commitments and UK net-zero policy create long-term uncertainty for field life assumptions.

Priority Exploration Targets

Given the above, we recommend focusing on plays where subsurface risk is balanced by high reward and existing infrastructure. Priority targets include:

- **Permo-Carboniferous plays near existing infrastructure:** e.g. Carboniferous reservoirs on the Mid-North Sea High and adjoining basins (e.g. Blocks 22/10–12, ~57°30'N 0°30'E) [28†L225-L233] . These use proven source rocks (Namurian coals/shales) and could tie back to nearby pipelines, reducing development cost.
- **East Irish Sea Permo-Triassic:** Unconventional leads like Pendleian sands under the St. George's Channel (e.g. near West Lancashire Basin) could yield gas/oil with modest wells. For instance, targets around 53°30'N 2°W (Cheshire Basin) leverage onshore data [60†L78-L85] .
- **Deep Triassic/Permian in Central NS:** Areas with high-quality Zechstein or Triassic reservoirs (e.g. tilted fault blocks on Edvard Graben or MNOSH flank) have significant gas resources; however, they require HPHT completions.
- **West of Shetland Paleocene fans:** Stratigraphic traps in the Foinaven/Schiehallion trend – e.g. deeper fan lobes west of Schiehallion (approx. 60°N 6°W) – where quality is proven [7†L1494-L1502] . These targets are remote but on established platforms.
- **Onshore structural leads:** Mature fields like Wytch Farm have limited upside, but smaller fault-block traps (e.g. Wessex sub-basins north of Salisbury, or Weald inversion anticlines) remain moderately attractive with low well costs.

Each target must be evaluated for geologic risk (fault complexity, pressure) versus infrastructure proximity. A prioritization flow (see diagram) can guide decision-making.

Data Gaps and Assumptions

The above assessment assumes access to public data (no proprietary seismic). Key data gaps include high-quality 3D seismic in underexplored areas (some West of Shetland and Irish Sea regions lack dense coverage). Subsurface uncertainties (depth mapping under basins, fault seal quality) remain in frontier plays. Economic assumptions (oil price, fiscal terms) are not specified. Finally, all recommendations assume drilling is permitted – the current UK ban on new licenses is a fundamental policy uncertainty.

Sources: UK government and industry data (NSTA/OGA, BGS, EITI) and peer-reviewed literature were used above [70†L199-L202] [28†L225-L233] [29†L248-L256] [57†L85-L88] [7†L1494-L1502] [45†L211-L219] [52†L97-L100] [49†L197-L205] . Integrated mapping studies (e.g. 21CXR [28†L269-L274]) provided basin insights. Any field statistics not cited are from standard industry databases (OGA Energy Portal). All tables/diagrams are schematic and for illustration; for operational planning, detailed data/QC per field is needed.
